

Few-Shot Personalization of Self-Supervised Wearable Biosignal Models

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Introduction

Key Contributions

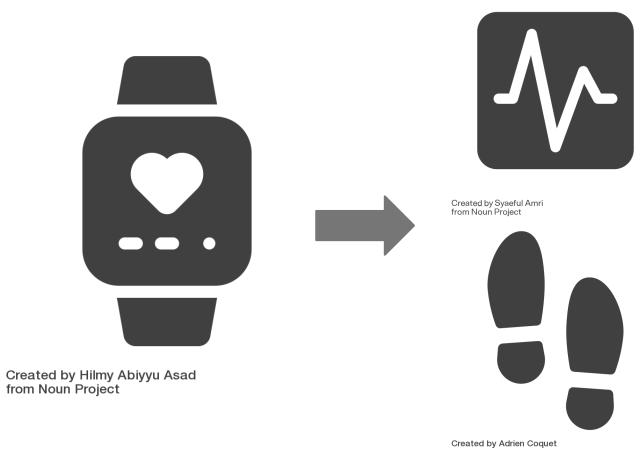
We demonstrate the effectiveness of state-space-based deep learning architectures (S4) [1] for learning generalizable representations of wearable biosignals.

We pre-train on minute-level heart rate and step count data from TILES-2018 [2], a longitudinal, naturalistic dataset consisting of physiological and behavioral signals collected from hospital workers.

Our approach largely matches or outperforms traditional supervised approaches on downstream objective and subjective classification tasks. We also show that the learned embeddings are able to perform well in subject-specific classification and forecasting tasks.

Background

Biosignal Representation Learning



Self-supervised learning (SSL) is a powerful approach for learning robust representations of data without requiring explicit labels.

Recent large-scale datasets have enabled the development of wearable sensor-based foundation models that are successful in downstream tasks such as classifying human activity recognition (HAR), demographics, and clinical diagnoses [3].

We extend previous work by focusing on capturing temporal patterns when learning representations.

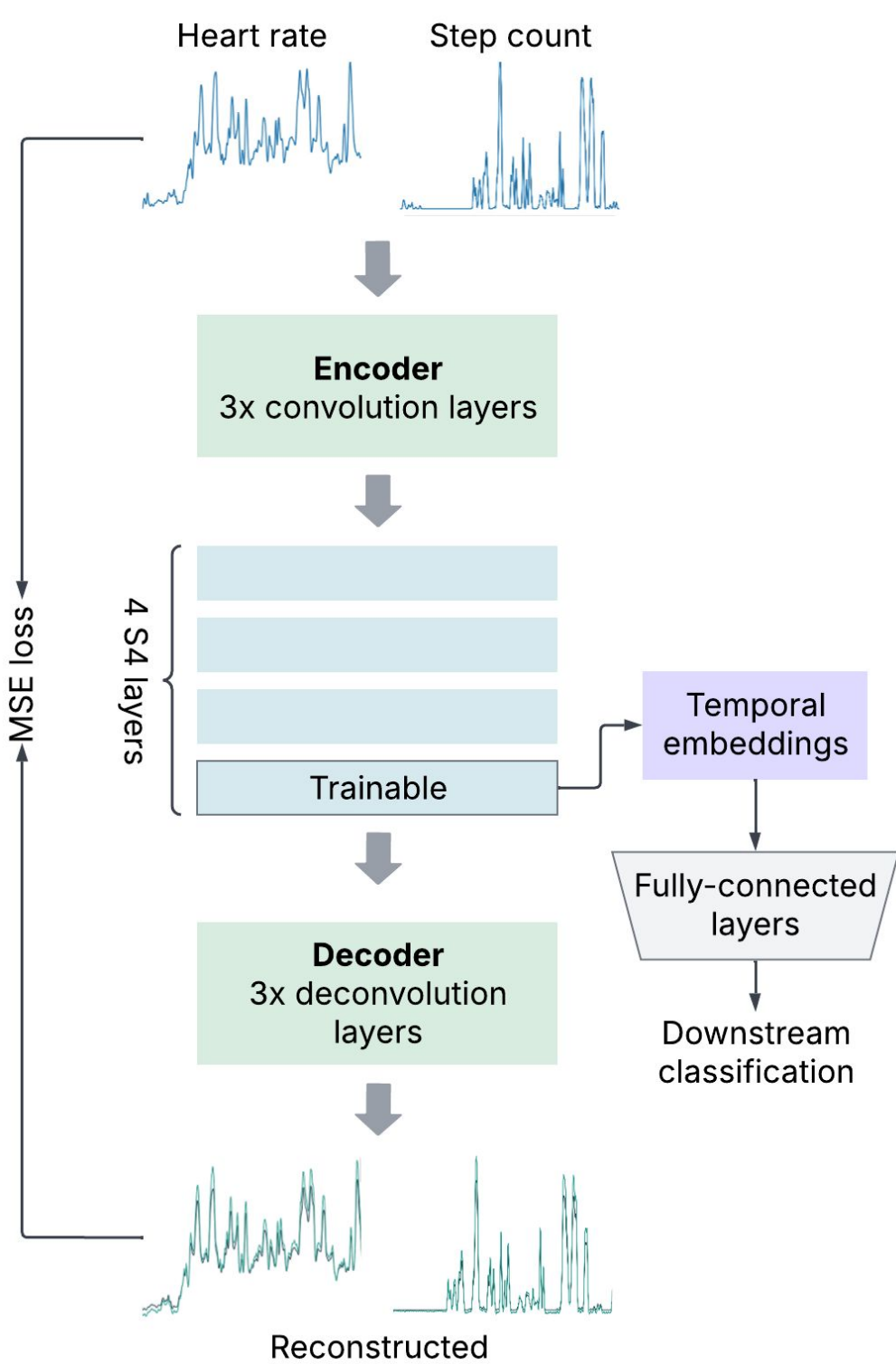
Datasets

TILES-2018 (open dataset): Longitudinal, naturalistic biosignal, behavioral, and environmental data collected from hospital workers over 10 weeks.

We use heart rate and step count from the Fitbit Charge 2 for pre-training and evaluation. Downstream labels were obtained from pre-study surveys and daily ecological momentary assessments.

Methodology

Self-Supervised Representation Learning via S4



The models were pre-trained with a batch size of 32 across 50 epochs with an AdamW optimizer with a base learning rate of 5e-4 and a weight decay of 1e-4. We pre-trained several models across a range of masking ratios: 0.1, 0.3, and 0.5.

We evaluated the generalizability of learned embeddings using a linear probe (LP) and fine-tuning with a convolutional classification head (FT).

Evaluation

Downstream Evaluation

Evaluation was carried out using the TILES-2018 held-out dataset.

- Shift prediction (day vs. night)
- Age prediction (≥ 40 vs. < 40)
- Daily self-reported stress prediction (high vs. low)
- Daily self-reported anxiety prediction (high vs. low)

The last S4 layer was unfrozen for fine-tuning. A classification head consisting of fully-connected layers was trained using 5-fold group cross-validation.

Method	Age				Shift			
	bACC	ACC	F1	AUC	bACC	ACC	F1	AUC
ST	SVM	0.653	0.500	0.000	0.500	0.792	0.500	0.000
	CNN	0.666	0.551	0.524	0.618	0.860	0.719	0.751
	S4	0.653	0.581	0.581	0.622	0.838	0.697	0.721
S4 MAE	10% LP	0.615	0.583	0.581	0.616	0.842	0.706	0.730
	10% FT	0.643	0.585	0.587	0.629	0.842	0.706	0.730
	30% LP	0.617	0.580	0.578	0.614	0.848	0.721	0.743
	30% FT	0.654	0.574	0.572	0.625	0.863	0.728	0.760
	50% LP	0.607	0.580	0.575	0.615	0.847	0.729	0.747
	50% FT	0.652	0.557	0.547	0.614	0.857	0.716	0.832

Method	Anxiety				Stress			
	bACC	ACC	F1	AUC	bACC	ACC	F1	AUC
ST	SVM	0.721	0.500	0.000	0.500	0.652	0.500	0.001
	CNN	0.718	0.503	0.434	0.542	0.639	0.520	0.482
	S4	0.651	0.511	0.505	0.528	0.604	0.532	0.529
S4 MAE	10% LP	0.603	0.519	0.517	0.528	0.580	0.526	0.525
	10% FT	0.603	0.519	0.517	0.528	0.580	0.526	0.525
	30% LP	0.638	0.522	0.521	0.533	0.588	0.529	0.528
	30% FT	0.687	0.514	0.492	0.542	0.610	0.530	0.524
	50% LP	0.648	0.519	0.516	0.533	0.599	0.531	0.528
	50% FT	0.707	0.505	0.455	0.535	0.629	0.528	0.509

Discussion

Importance of Capturing Temporal Patterns

Results suggest that capturing temporal patterns within days can improve the generalizability of learned representations of wearable biosignals.

Importantly, we evaluate the ability of our model's learned representations to capture temporal dynamics related to self-reported stress. These subjective tasks are more difficult due to high inter- and intra-subject variability.

Future Work

- Evaluation of cross-dataset generalizability (All of Us) on a wider variety of labels.
- Learning representations across longer time scales (day/week level)
- Incorporating within-person physiological dynamics.

References

[1] Gu, A., Goel, K., & Ré, C. Efficiently modeling long sequences with structured state spaces. arXiv preprint arXiv:2111.00396 (2021).

[2] Mundnich, K. et al. Tiles-2018, a longitudinal physiologic and behavioral dataset of hospital workers. Scientific Data 7 (2020).

[3] Xu, M. et al. "Lsm-2: Learning from incomplete wearable sensor data," 2025.

[4] Gu, A, et al. "Combining recurrent, convolutional, and continuous-time models with linear state space layers." *Advances in neural information processing systems* 34 (2021): 572-585.